1. System Architecture Overview

High-level architecture:

```
        USERS
    (Web + Mobile App)
          |
          ▼
      CDN + Load Balancer
          |
          ▼
        API Gateway
          |
   ┌──────────┼──────────┐
   ▼      ▼      ▼
Auth Service  User Service  Market Service
                  |
                  ▼
            Market Data APIs
        (Polygon / Finnhub / AlphaVantage)

   ▼      ▼      ▼
Event Engine  Prediction Engine  Board Service
   |        |        |
   ▼        ▼        ▼
NLP Pipeline  ML Prediction   User Content

          |
          ▼
      Message Queue (Kafka)
          |
          ▼
```

```
Data Processing Layer
      |
      ▼
 Core Databases
      |
      ▼
Analytics / Search
```

2. Frontend Layer

Two clients share the same API.

Web Application

Tech stack:

```
Next.js
TypeScript
TailwindCSS
ShadCN UI
React Query
Recharts
```

Features:

- event dashboard
- market panels
- intelligence boards
- asset analytics
- global map visualization

Mobile Application

Tech stack:

```
React Native
Expo
TypeScript
```

Features:

- real-time alerts

- event feed
- watchlists
- board management

---

3. API Gateway

Single entry point for all requests.

Tools:

```
NGINX
Kong
AWS API Gateway
```

Responsibilities:

```
rate limiting
authentication
routing
request logging
```

---

4. Backend Microservices

The backend is divided into **domain services**.

---

1. User Service

Handles:

```
accounts
authentication
preferences
subscriptions
```

Tech:

```
FastAPI
PostgreSQL
JWT
```

---

2. Event Intelligence Service

Processes news and extracts structured geopolitical events.

Pipeline:

```
News ingestion
↓
NLP extraction
↓
Event classification
↓
Impact tagging
```

Tech:

```
Python
spaCy
Transformers
FastAPI
```

Example event:

```
{
 event: "China bans rare earth exports",
 region: "China",
 sector: "EV supply chain",
 assets: ["TSLA","BYD"],
 impact: "negative"
}
```

---

3. Market Data Service

Handles:

```
live prices
market metadata
historical data
```

APIs used:

```
Finnhub
Polygon
Alpha Vantage
Yahoo Finance
```

Cached via:

```
Redis
```

---

4. Prediction Service

AI system that predicts **market reaction to events.**

Inputs:

```
news sentiment
historical market reaction
macro indicators
```

Models:

```
LSTM
Transformer sentiment model
Gradient boosting
```

Frameworks:

```
PyTorch
scikit-learn
HuggingFace
```

---

5. Board / Knowledge Graph Service

Handles:

```
Pinterest-style boards
pins
collections
shared insights
```

Example:

```
Board: US–China Tech War
Pins:
• Nvidia export ban
• Semiconductor supply chain
• TSMC market reaction
```

---

5. Data Ingestion Pipeline

This is one of the most important systems.

Architecture:

```
News APIs
RSS Feeds
Think Tank Reports
Government Releases
```

Pipeline:

```
Ingestion Worker
↓
Kafka Event Stream
↓
NLP Processing
```

```
↓
Event Classification
↓
Market Mapping
↓
Prediction Engine
↓
Database
```

Tools:

```
Apache Kafka
Celery
Airflow
```

---

6. Message Queue (Real-Time Event Processing)

Kafka handles **stream processing**.

Use cases:

```
news ingestion
event extraction
prediction updates
alert triggers
```

Example pipeline:

```
News Article
→ Kafka topic
→ NLP Worker
→ Event Classification
→ Market Mapping
→ Prediction Engine
```

---

7. Databases

Use **polyglot persistence**.

---

1. Primary Database

```
PostgreSQL
```

Stores:

```
users
events
assets
boards
predictions
```

---

2. Time-Series Database

```
TimescaleDB
```

Stores:

```
market prices
volume
historical indicators
```

---

3. Cache Layer

```
Redis
```

Stores:

```
market prices
API results
session tokens
```

---

4. Search Engine

```
Elasticsearch
```

Used for:

```
news search
event filtering
asset search
```

---

8. AI / ML Infrastructure

The ML stack is separate from backend.

Architecture:

```
Prediction API
↓
ML Model Server
↓
Feature Store
```

```
↓
Training Pipelines
```

Frameworks:

```
PyTorch
MLflow
HuggingFace
```

Deployment:

```
TorchServe
or
FastAPI model server
```

---

9. Infrastructure (Cloud)

Recommended provider:

```
AWS
```

Services:

```
EC2
EKS (Kubernetes)
S3
CloudFront
RDS
ElastiCache
MSK (Kafka)
```

---

10. DevOps Pipeline

CI/CD pipeline:

```
GitHub
↓
GitHub Actions
↓
Docker Build
↓
Kubernetes Deployment
```

Infrastructure management:

```
Terraform
```

---

11. Security Layer

Essential security measures.

Authentication:

```
JWT
OAuth
Auth0
```

Protection:

```
API rate limiting
DDoS protection
input validation
RBAC
```

---

12. Notification System

Alerts triggered when events affect tracked assets.

Example:

```
Event detected:
China restricts chip exports

User tracking:
NVDA

Notification:
High volatility expected
```

Channels:

```
mobile push
email
web notifications
```

Tools:

```
Firebase
AWS SNS
```

---

13. Scalability Strategy

System designed for growth.

Horizontal scaling:

```
Kubernetes auto-scaling
```

Caching:

```
Redis
```

Event streaming:

```
Kafka partitions
```

---

14. Observability

Monitoring tools:

```
Prometheus
Grafana
ELK stack
```

Tracks:

```
API latency
ML model performance
pipeline health
```

---

15. Development Phases

Phase 1 — MVP (2 months)

Build:

```
news feed
event classification
market dashboard
```

Phase 2 — Intelligence Engine

Add:

```
AI event extraction
impact mapping
prediction models
```

Phase 3 — Platform Features

Add:

```
boards
alerts
mobile app
```

---

16. Estimated System Scale

Target scale:

```
100k users
10k daily events
100M market records
```

Architecture supports:

```
millions of events
real-time data
```

---

17. Strategic Insight

The **true moat** of this product is **data intelligence**, not UI.

Your competitive advantage will be:

```
Event → Market Mapping Dataset
```

Over time this becomes a **proprietary geopolitical intelligence dataset**.

That is extremely valuable.